$python3.12 ~/icarus.py

ICARUS

Integrated Cosmic Analysis Research Utility for Spectroscopy

Version: 3.0.0 | Author: Joseph Havens

➤ Initializing Research Environment ...

[INFO] Config validation successful.

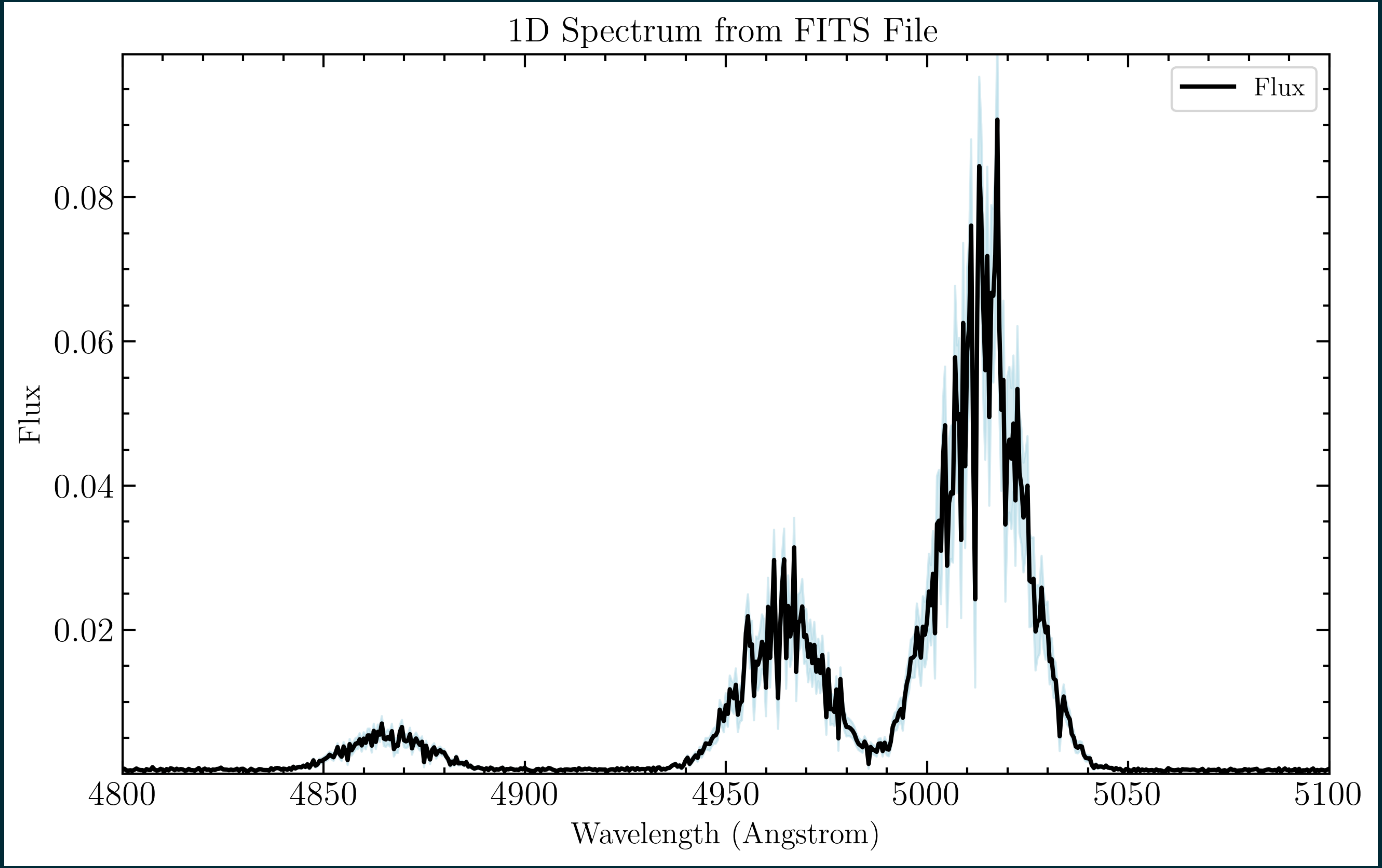
➤ Registering Survey Catalog…

[INFO] Successfully loaded 1 presentation from ~/Astr_Seminar/

University of Kansas | Dr. Allison Kirkpatrick | grad: May 2027 | josephhavens.me

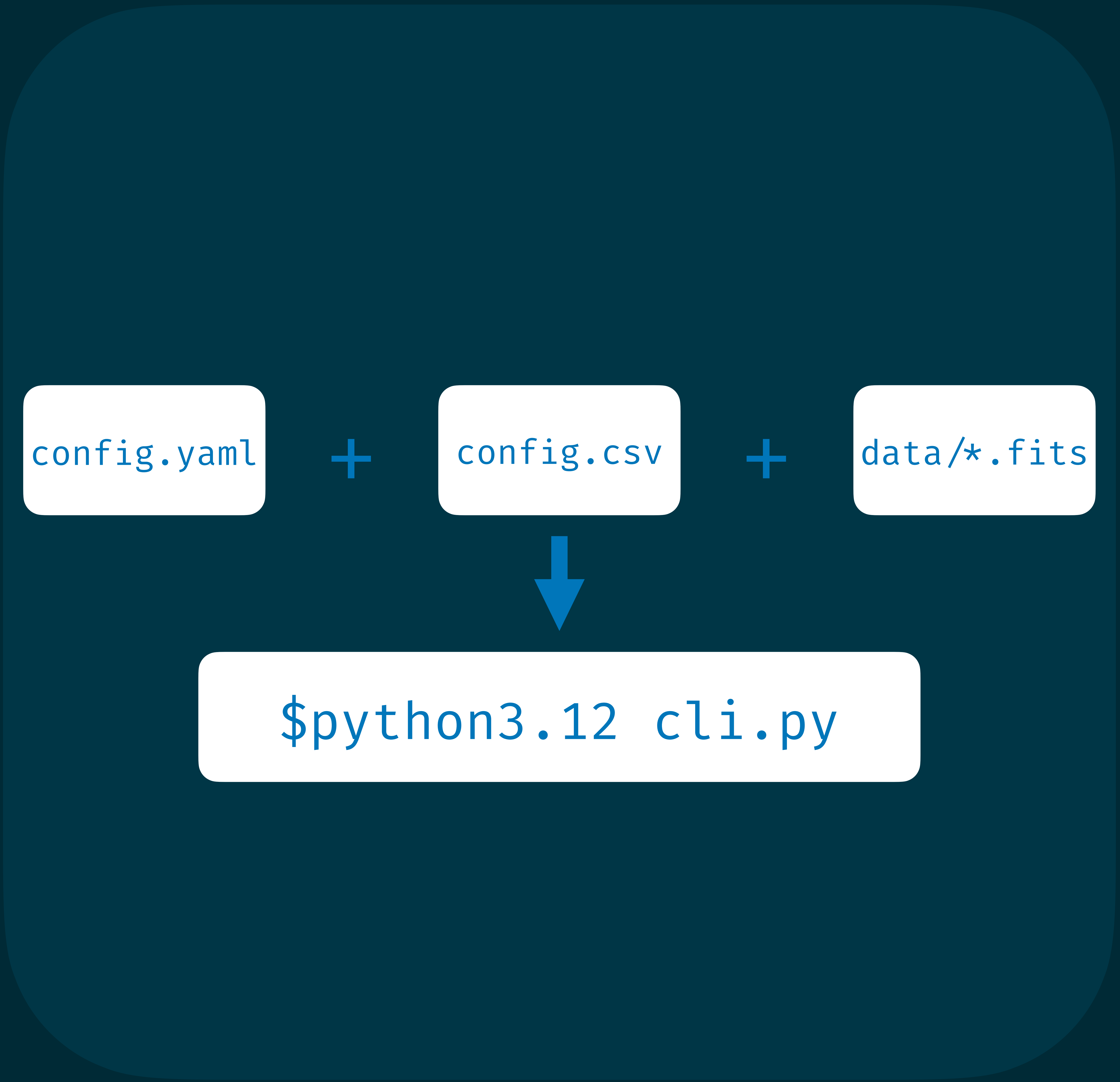
The Problem

- Need a reliable Emission Line fitting pipeline with AGN-capabilities
  - Broad line
  - Narrow line
  - blended line analysis
  - Upper-limit and low SNR fitting


1D Spectrum from FITS File
Flux
0.08
0.06
0.04
0.02
4800
4850
4900
4950
5000
5050
5100
Wavelength (Angstrom)
Flux

The Layout

```
├── cli.py
├── datastructures.py
├── exceptions.py
├── fitting_engine.py
├── ingest.py
├── __init__.py
├── logger.py
├── main.py
├── matplotlibparams.py
├── models.py
├── plotting.py
└── science_metrics.py
```



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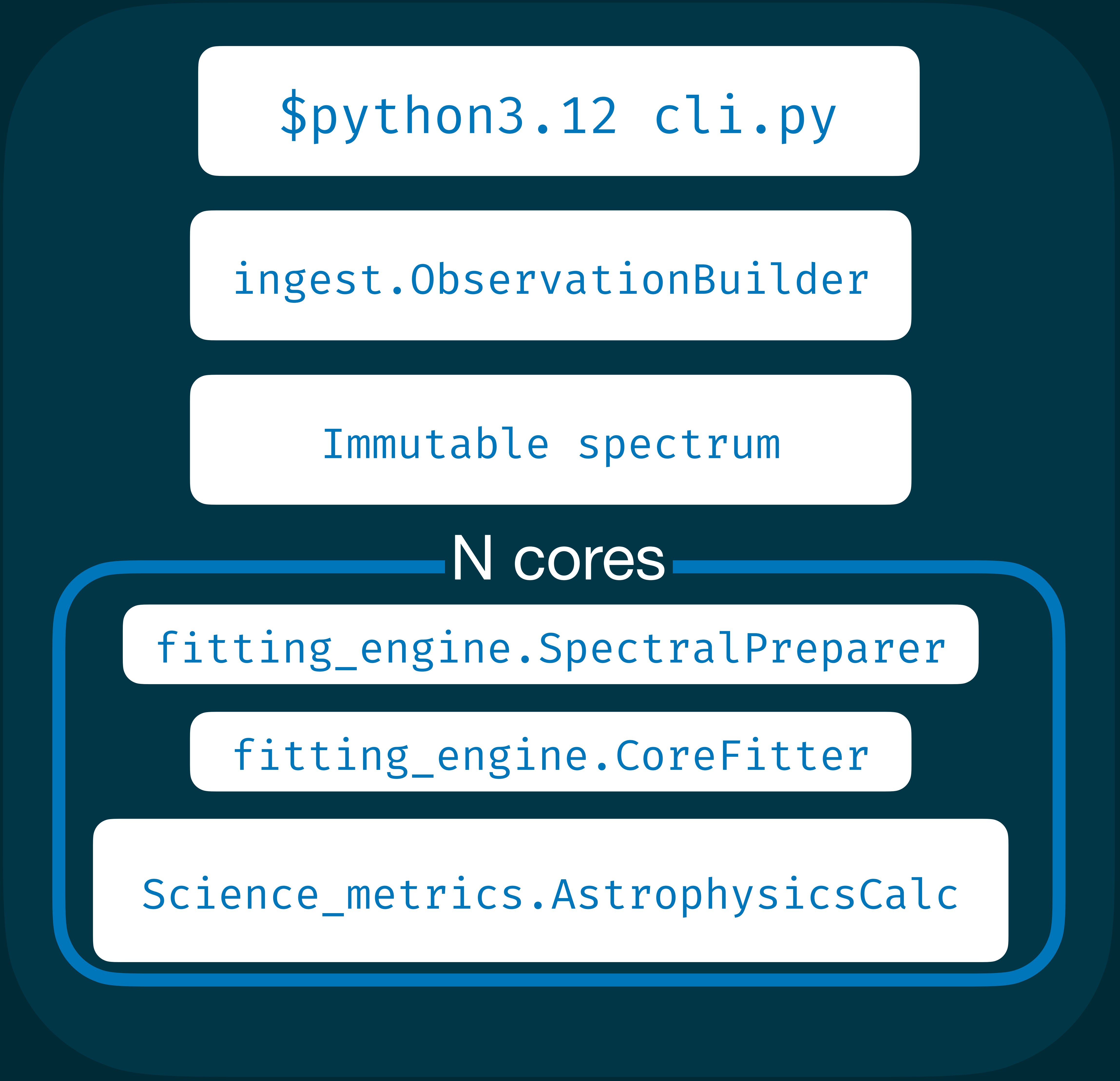
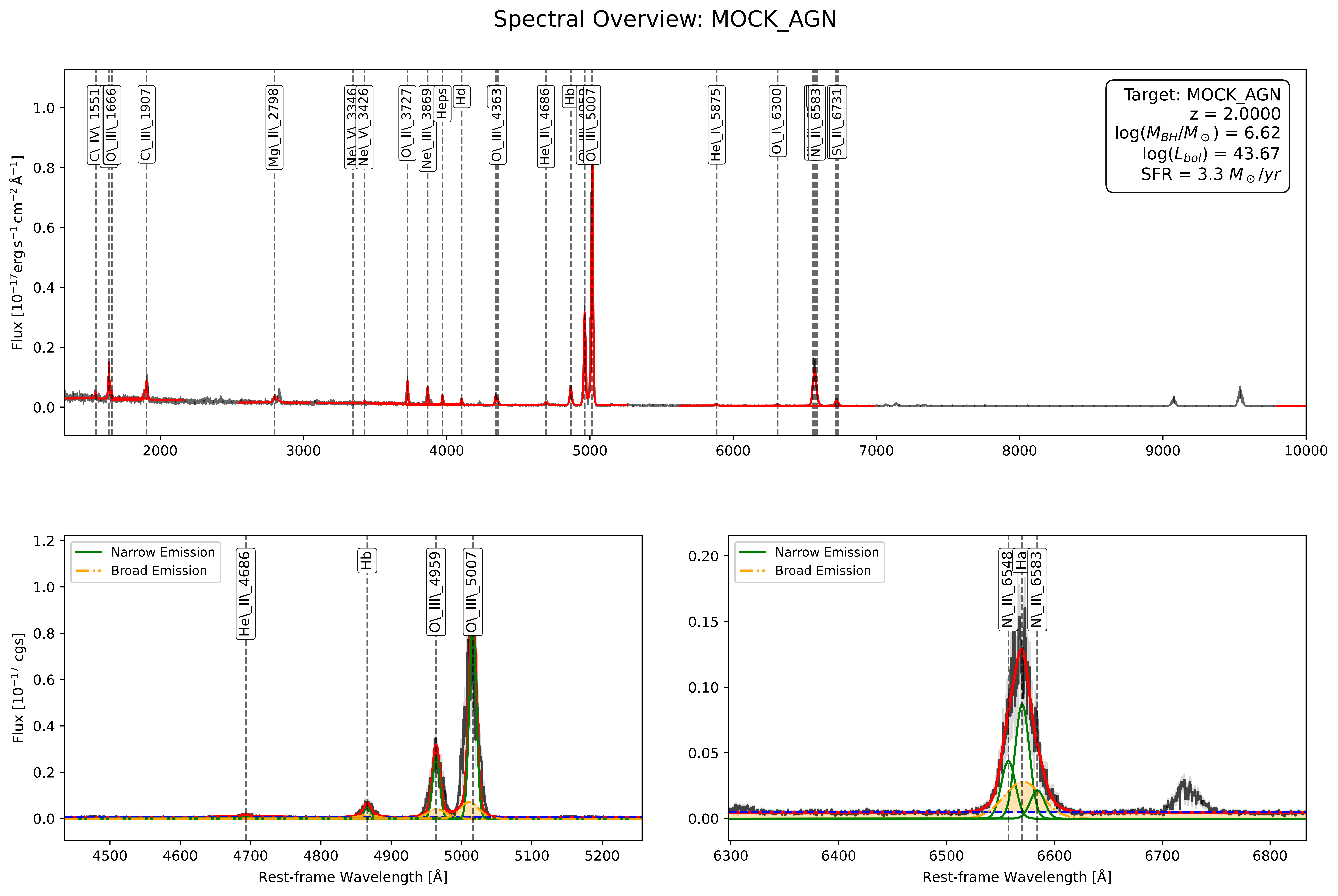


Outputs



Spectral Overview: MOCK_AGN
Target: MOCK_AGN
z = 2.0000
log($M_{BH}/M_\odot$) = 6.62
log(L_{bol}) = 43.67
SFR = 3.3 $M_\odot/yr$
Flux [10^{-17} erg s^{-1} cm^{-2} Å^{-1}]
C_IV_1551
O_III_1666
C_III_1907
Mg_II_2798
Ne_V_3346
Ne_V_3426
O_II_3727
Ne_III_3869
Heps
Hd
O_III_4363
He_II_4686
Hb
O_III_5007
He_I_5875
O_I_6300
N_II_6583
S_II_6731
Narrow Emission
Broad Emission
He_II_4686
Hb
O_III_4959
O_III_5007
Flux [10^{-17} cgs]
Rest-frame Wavelength [Å]
N_II_6548
Ha
N_II_6583
Rest-frame Wavelength [Å]

Funny error

Target: MOCK_AGN

$z = 2.0000$

$\log(M_{BH}/M_\odot) = 15.42$

$\log(L_{bol}) = 59.67$

$\mathrm{SFR} = 32655929825105664.0\ M_\odot/yr$

Future updates

- -Current Version: 3.0.0
- -Using “pride versioning”
  - -major.minor.shame
- -future updates:
  - -Color and window fixes
  - -“Boxcar Filter”
  - -OIII blue wing gaussian
  - -Upper Limit
  - -consider scaling Broad Lines

- -validation needed:
  - -more CLOUDY tests
  - -test when it breaks

Thanks and Acknowledgement

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